Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 14: Two-Port Circuits

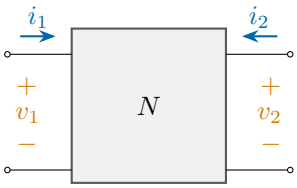
M. Mert Ankaralı

Colour code used in these notes and in the slides

■ current i ■ voltage v, node voltage e

Motivation. Previously, a one-port without independent sources was described by one number, its input resistance: $v = R_{\text{in}}\, i$. Many circuits have an input and an output: an amplifier, a transformer, a filter. Such a circuit has two ports, and its description is the natural generalization of $v = R\, i$: two port voltages, two port currents and a 2×2 matrix. This lecture introduces four such matrices (**resistance**, **conductance**, **hybrid** and **transmission**) and the **power** of a two-port.

1. Two-port circuits



N is an LTI resistive circuit with no independent sources. At each port the current enters at the + terminal and leaves at the − terminal. There are four port variables v_1, i_1, v_2, i_2 and two equations between them: two of the variables can be written in terms of the other two.

given	find	matrix	equation
i_1, i_2	v_1, v_2	resistance $\mathbf{R}$	$[\, v_1 \;\; v_2 \,]^T = \mathbf{R}\, [\, i_1 \;\; i_2 \,]^T$
v_1, v_2	i_1, i_2	conductance $\mathbf{G}$	$[\, i_1 \;\; i_2 \,]^T = \mathbf{G}\, [\, v_1 \;\; v_2 \,]^T$
i_1, v_2	v_1, i_2	hybrid $\mathbf{H}$	$[\, v_1 \;\; i_2 \,]^T = \mathbf{H}\, [\, i_1 \;\; v_2 \,]^T$
v_1, i_2	i_1, v_2	hybrid $\mathbf{H}'$	$[\, i_1 \;\; v_2 \,]^T = \mathbf{H}'\, [\, v_1 \;\; i_2 \,]^T$
v_2, $-i_2$	v_1, i_1	transmission $\mathbf{T}$	$[\, v_1 \;\; i_1 \,]^T = \mathbf{T}\, [\, v_2 \;\; -i_2 \,]^T$

2. The resistance matrix

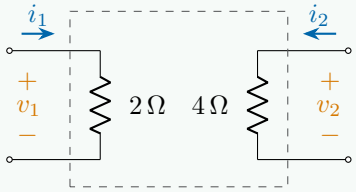
Resistance matrix

The port voltages in terms of the port currents:

$$v_1 = r_{11}\, i_1 + r_{12}\, i_2$$
$$v_2 = r_{21}\, i_1 + r_{22}\, i_2$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

Example 14.1 — two separate resistors



The two-port is a $2\,\Omega$ resistor across port 1 and a $4\,\Omega$ resistor across port 2. Find $\mathbf{R}$.

Solution

Step 1. Each port alone

Each resistor carries only its own port current:

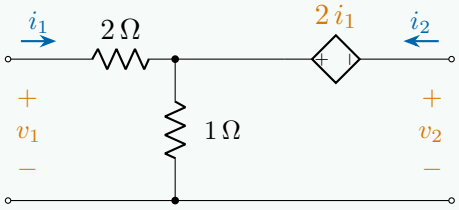
$$v_1 = 2\,i_1 + 0 \cdot i_2, \qquad v_2 = 0 \cdot i_1 + 4\,i_2.$$

Step 2. The matrix

$$\mathbf{R} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \Omega.$$

The off-diagonal entries are zero: the two ports are not coupled.

Example 14.2 — a dependent source inside



The two-port contains a $2\,\Omega$ and a $1\,\Omega$ resistor and a current-controlled voltage source $2\,i_1$. Find $\mathbf{R}$.

Solution

Step 1. The current of $1\,\Omega$

By KCL at the middle node, the $1\,\Omega$ resistor carries $i_1 + i_2$ downwards.

Step 2. KVL from each port

From port 1 through $2\,\Omega$ and $1\,\Omega$:

$$v_1 = 2\,i_1 + (i_1 + i_2) = 3\,i_1 + i_2.$$

From port 2 through the dependent source and $1\,\Omega$:

$$v_2 = -2\,i_1 + (i_1 + i_2) = -i_1 + i_2.$$

Step 3. The matrix

Collect the terms in i_1 and i_2:

$$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \Omega.$$

Without the dependent source $\mathbf{R}$ would be symmetric; here $r_{12} \neq r_{21}$.

2.1. The entries from open-circuit tests

The matrix equation in the box above also tells how to find $\mathbf{R}$ one column at a time: make one port current zero, that is, leave that port **open**.

Port 2 open ($i_2 = 0$). Drive port 1 with a current i_1, for example with a current source. The second column of $\mathbf{R}$ is multiplied by zero and drops out:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} i_1 \\ 0 \end{bmatrix} = \begin{bmatrix} r_{11}\,i_1 \\ r_{21}\,i_1 \end{bmatrix}.$$

Both port voltages are proportional to i_1. Dividing by i_1 gives the **first column**:

$$r_{11} = \left.\frac{v_1}{i_1}\right|_{i_2=0}, \qquad r_{21} = \left.\frac{v_2}{i_1}\right|_{i_2=0}.$$

Port 1 open ($i_1 = 0$). Drive port 2 with a current i_2. Now the first column drops out:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \begin{bmatrix} 0 \\ i_2 \end{bmatrix} = \begin{bmatrix} r_{12}\,i_2 \\ r_{22}\,i_2 \end{bmatrix},$$

and dividing by i_2 gives the **second column**:

$$r_{12} = \left.\frac{v_1}{i_2}\right|_{i_1=0}, \qquad r_{22} = \left.\frac{v_2}{i_2}\right|_{i_1=0}.$$

What the entries mean. The diagonal entries are input resistances: r_{11} is the resistance seen at port 1 when port 2 is open, and r_{22} is the resistance seen at port 2 when port 1 is open. The off-diagonal entries are transfer resistances: r_{21} is the voltage that appears at the open port 2 per ampere into port 1, and r_{12} the other

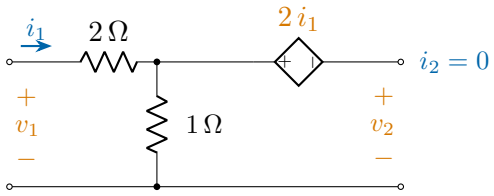
way round. Since both tests leave a port open, **R** is also called the **open-circuit resistance matrix**. Each test is an ordinary one-source linear circuit.

Example 14.2 revisited — open-circuit tests

Find **R** of Example 14.2 again, this time from the two open-circuit tests.

Solution

Step 1. Port 2 open: $i_2 = 0$

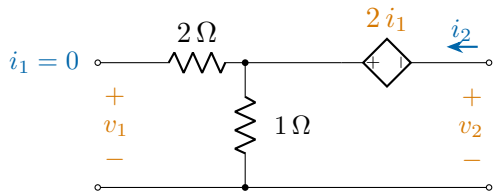


No current flows through the dependent source, so v_2 is the voltage of 1 Ω minus the dependent source:

$$v_1 = (2+1)\, i_1 = 3\, i_1$$
$$v_2 = i_1 - 2\, i_1 = -i_1$$

So $r_{11} = 3\,\Omega$ and $r_{21} = -1\,\Omega$.

Step 2. Port 1 open: $i_1 = 0$



No current flows in 2 Ω, and the dependent source is $2\, i_1 = 0$:

$$v_1 = i_2$$
$$v_2 = i_2$$

So $r_{12} = 1\,\Omega$ and $r_{22} = 1\,\Omega$: the same **R** as before. ✓

3. The conductance matrix

Conductance matrix

The port currents in terms of the port voltages:

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}}_{\mathbf{G}} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

The entries follow from **short-circuit tests**:

$$g_{11} = \left.\frac{i_1}{v_1}\right|_{v_2=0}, \quad g_{21} = \left.\frac{i_2}{v_1}\right|_{v_2=0},$$
$$g_{12} = \left.\frac{i_1}{v_2}\right|_{v_1=0}, \quad g_{22} = \left.\frac{i_2}{v_2}\right|_{v_1=0}.$$

The entries, column by column. Now make one port *voltage* zero, that is, **short** that port, and drive the other port with a voltage.

Port 2 shorted ($v_2 = 0$):

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = \begin{bmatrix} g_{11}\, v_1 \\ g_{21}\, v_1 \end{bmatrix}$$

the **first column**.

Port 1 shorted ($v_1 = 0$):

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} g_{12}\, v_2 \\ g_{22}\, v_2 \end{bmatrix}$$

the **second column**.

What the entries mean. The diagonal entries are input conductances: g_{11} is the conductance seen at port 1 when port 2 is shorted, and g_{22} the conductance seen at port 2 when port 1 is shorted. The off-diagonal entries are transfer conductances: g_{21} is the current that flows in the shorted port 2 per volt at port 1, and g_{12} the other way round. Since both tests short a port, $\mathbf{G}$ is also called the **short-circuit conductance matrix**.

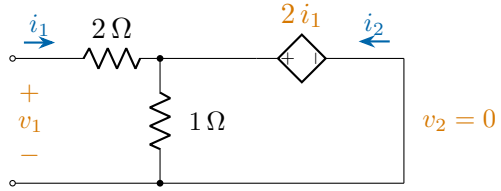
If $\mathbf{R}$ is invertible, $\mathbf{v} = \mathbf{R}\,\mathbf{i}$ gives $\mathbf{i} = \mathbf{R}^{-1}\mathbf{v}$, so $\mathbf{G} = \mathbf{R}^{-1}$ (and $\mathbf{R} = \mathbf{G}^{-1}$). Note that $g_{11} \neq 1/r_{11}$ in general: the two are measured with the other port shorted and open, respectively.

Example 14.3 — the conductance matrix of Example 14.2

Find the first column of $\mathbf{G}$ for the two-port of Example 14.2 by a short-circuit test. Then find all of $\mathbf{G}$.

Solution

Step 1. Port 2 shorted: $v_2 = 0$



Two meshes, with i_1 and i_2 as the mesh currents.

$$\text{Mesh 1:} \quad -v_1 + 2\,i_1 + (i_1 + i_2) = 0$$
$$\text{Mesh 2:} \quad -2\,i_1 + (i_1 + i_2) = 0$$

Step 2. Solve

Mesh 2 gives $i_2 = i_1$. Mesh 1 then gives $v_1 = 2\,i_1 + 2\,i_1 = 4\,i_1$.

Step 3. The first column

$$g_{11} = \left.\frac{i_1}{v_1}\right|_{v_2=0} = \frac{1}{4}\,\text{S}, \qquad g_{21} = \left.\frac{i_2}{v_1}\right|_{v_2=0} = \frac{1}{4}\,\text{S}.$$

Step 4. All of G

The second test ($v_1 = 0$) gives the second column in the same way. The result is the inverse of $\mathbf{R}$:

$$\mathbf{G} = \mathbf{R}^{-1} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}^{-1} = \frac{1}{4}\begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} \text{S}.$$

Check: $\mathbf{G}\,\mathbf{R} = \frac{1}{4}\begin{bmatrix} 3+1 & 1-1 \\ 3-3 & 1+3 \end{bmatrix} = \mathbf{1}$. ✓

Note that $g_{11} = \frac{1}{4}\,\text{S} \neq 1/r_{11} = \frac{1}{3}\,\text{S}$.

Remark

$\mathbf{R}$ or $\mathbf{G}$ need not exist. When both exist, $\mathbf{G} = \mathbf{R}^{-1}$. The ideal transformer (Example 14.5) has neither.

4. Hybrid matrices

Hybrid matrices

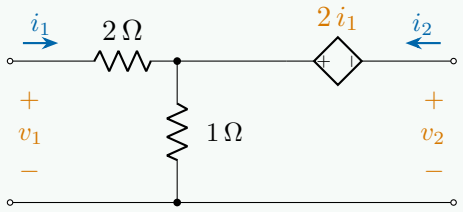
Mix one current and one voltage on each side:

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \underbrace{\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}, \qquad \begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \underbrace{\begin{bmatrix} h'_{11} & h'_{12} \\ h'_{21} & h'_{22} \end{bmatrix}}_{\mathbf{H}'} \begin{bmatrix} v_1 \\ i_2 \end{bmatrix},$$

with $\mathbf{H}' = \mathbf{H}^{-1}$ when the inverse exists.

The entries of $\mathbf{H}$ follow from two tests: port 2 shorted ($v_2 = 0$) gives $h_{11} = v_1/i_1$ and $h_{21} = i_2/i_1$; port 1 open ($i_1 = 0$) gives $h_{12} = v_1/v_2$ and $h_{22} = i_2/v_2$. The entries have different units: h_{11} is a resistance, h_{22} a conductance, h_{12} and h_{21} have no units.

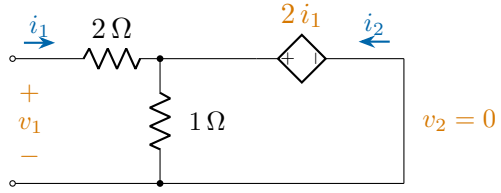
Example 14.4 — the hybrid matrix of a circuit



This is the two-port of Example 14.2. Find $\mathbf{H}$.

Solution

Step 1. Port 2 shorted: $v_2 = 0$



Mesh 2 contains the dependent source and 1 Ω:

$$-2\,i_1 + (i_1 + i_2) = 0$$

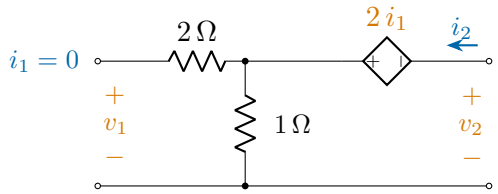
$$\Rightarrow \; i_2 = i_1.$$

Mesh 1 then gives

$$v_1 = 2\,i_1 + (i_1 + i_2) = 4\,i_1.$$

So $h_{11} = 4\,\Omega$ and $h_{21} = 1$.

Step 2. Port 1 open: $i_1 = 0$



The dependent source is $2\,i_1 = 0$, a short circuit, and no current flows in 2 Ω:

$$v_2 = 1 \cdot i_2 \;\Rightarrow\; i_2 = v_2.$$

The voltage of port 1 is the voltage of 1 Ω:

$$v_1 = i_2 = v_2.$$

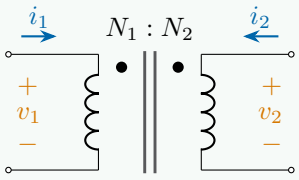
So $h_{12} = 1$ and $h_{22} = 1\,\mathrm{S}$.

Step 3. The matrix

$$\mathbf{H} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}.$$

Check: Example 14.2 gives $\mathbf{R} = \left[\begin{smallmatrix} 3 & 1 \\ -1 & 1 \end{smallmatrix}\right]$. Solving its second row for i_2 and substituting into the first row gives the same $\mathbf{H}$. ✓

Example 14.5 — the hybrid matrix of the ideal transformer



The ideal transformer $N_1 : N_2$ (previously) satisfies

$$\frac{v_1}{N_1} = \frac{v_2}{N_2}, \qquad N_1\,i_1 + N_2\,i_2 = 0.$$

Find $\mathbf{H}$. Does $\mathbf{R}$ exist?

Solution

Step 1. Solve for v_1 and i_2

With the turns ratio $n = N_1/N_2$:

$$v_1 = n\,v_2, \qquad i_2 = -n\,i_1.$$

Step 2. The matrix

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix} \qquad \Rightarrow \qquad \mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}.$$

Step 3. No resistance matrix

With both ports open, $i_1 = i_2 = 0$, but $v_1 = n\,v_2$ can take any value. So the voltages are not determined by the currents: **R** does not exist. In the same way, shorting both ports ($v_1 = v_2 = 0$) leaves $i_1 = -i_2/n$ free, so **G** does not exist either.

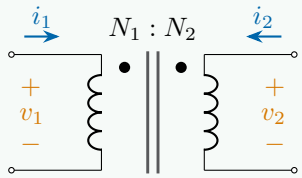
5. The transmission matrix

Transmission matrix (chain parameters)

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$

Port 1 in terms of port 2. The minus sign makes $-i_2$ the current that leaves port 2, for example towards a load.

Example 14.6 — the transmission matrix of the ideal transformer



Find **T** for the ideal transformer $N_1 : N_2$.

Solution

Step 1. Port 1 in terms of port 2

From the two transformer equations:

$$v_1 = \frac{N_1}{N_2}\,v_2, \qquad i_1 = -\frac{N_2}{N_1}\,i_2 = \frac{N_2}{N_1}\,(-i_2).$$

Step 2. The matrix

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} N_1/N_2 & 0 \\ 0 & N_2/N_1 \end{bmatrix} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix} \quad \Rightarrow \quad \mathbf{T} = \begin{bmatrix} N_1/N_2 & 0 \\ 0 & N_2/N_1 \end{bmatrix}.$$

Which test? Open or short

Every matrix has two *inputs*: the variables in the vector on the right-hand side. To get one column, set the other input to zero. A zero current means that port is **open**; a zero voltage means that port is **shorted**. (This is the same as killing a source: a dead current source is an open, a dead voltage source is a short.)

Matrix	Inputs	1st column	2nd column
R: $[\,v_1\ v_2\,]^T = \mathbf{R}\,[\,i_1\ i_2\,]^T$	i_1, i_2	$i_2 = 0$: port 2 **open**	$i_1 = 0$: port 1 **open**
G: $[\,i_1\ i_2\,]^T = \mathbf{G}\,[\,v_1\ v_2\,]^T$	v_1, v_2	$v_2 = 0$: port 2 **shorted**	$v_1 = 0$: port 1 **shorted**
H: $[\,v_1\ i_2\,]^T = \mathbf{H}\,[\,i_1\ v_2\,]^T$	i_1, v_2	$v_2 = 0$: port 2 **shorted**	$i_1 = 0$: port 1 **open**
H′: $[\,i_1\ v_2\,]^T = \mathbf{H}'\,[\,v_1\ i_2\,]^T$	v_1, i_2	$i_2 = 0$: port 2 **open**	$v_1 = 0$: port 1 **shorted**
T: $[\,v_1\ i_1\,]^T = \mathbf{T}\,[\,v_2\ -i_2\,]^T$	v_2, $-i_2$	$i_2 = 0$: port 2 **open**	$v_2 = 0$: port 2 **shorted**

The names already say it: **R** is the open-circuit matrix and **G** the short-circuit matrix. **H** mixes a current and a voltage as inputs, so its tests are mixed too, and its entries have mixed units. **T** is the odd one: both tests are done at port 2, and port 1 is only measured. The port that is not zeroed is driven by a source of its own input type: a current source if its input is a current, a voltage source if it is a voltage.

6. Measuring a two-port

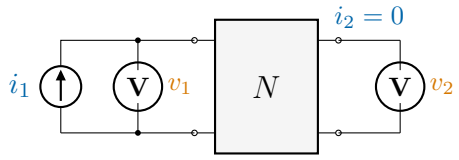
In a laboratory the two-port is a closed box with four terminals. Nothing inside is known, but if it is linear, **R** (or **G**, **H**, **T**) has only four numbers. Each experiment gives two equations, so **two experiments** are enough. The open- and short-circuit tests are the two experiments that give the entries most directly.

6.1. Ideal meters are opens and shorts

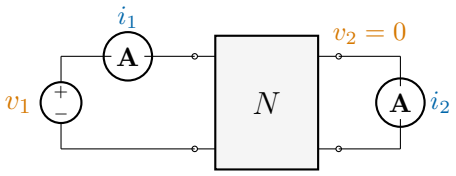
An ideal **voltmeter** draws no current: it is an *open* circuit. An ideal **ammeter** has no voltage across it: it is a *short* circuit. So

- a port with only a voltmeter across it is an **open** port, and the meter reads its voltage;
- a port closed by only an ammeter is a **shorted** port, and the meter reads its current.

The tests of this lecture are therefore real experiments: a source at one port, and meters at both ports.



Open-circuit test. A current source drives port 1; voltmeters read v_1 and v_2. Then $r_{11} = v_1/i_1$ and $r_{21} = v_2/i_1$.



Short-circuit test. A voltage source drives port 1; ammeters read i_1 and i_2. Then $g_{11} = i_1/v_1$ and $g_{21} = i_2/v_1$.

Each entry is one meter reading divided by the source value; no circuit analysis is needed. Moving the source to port 2 gives the second column. For **H**, port 2 gets an ammeter and port 1 a voltmeter, and so on, following the table above.

6.2. When a port cannot be opened or shorted

Opening or shorting a port is not always possible. A short at the output of an amplifier may damage it. An open-circuit test may be impossible: with port 2 of an ideal transformer open, no current can enter port 1, so its **R** does not exist. Then any **two independent experiments** still work. Put the two measured current vectors side by side, and the two voltage vectors side by side:

$$\underbrace{\begin{bmatrix} v_1^{(1)} & v_1^{(2)} \\ v_2^{(1)} & v_2^{(2)} \end{bmatrix}}_{\mathbf{V}} = \mathbf{R} \underbrace{\begin{bmatrix} i_1^{(1)} & i_1^{(2)} \\ i_2^{(1)} & i_2^{(2)} \end{bmatrix}}_{\mathbf{I}} \qquad \Rightarrow \qquad \mathbf{R} = \mathbf{V}\,\mathbf{I}^{-1}.$$

The superscript is the experiment number. The experiments are independent when **I** is invertible. The open-circuit tests are the special case **I** diagonal, where no inverse has to be computed.

Example 14.7 — R from two measurements

Two experiments on a linear two-port give

	i_1 (A)	i_2 (A)	v_1 (V)	v_2 (V)
Experiment 1	1	1	5	5
Experiment 2	1	−1	3	−1

Neither experiment has an open port. Find **R**.

Solution

Step 1. Stack the experiments

$$\begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} = \mathbf{R} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Step 2. Solve

$\det \mathbf{I} = -2 \neq 0$, so the experiments are independent, and

$$\mathbf{I}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{R} = \begin{bmatrix} 5 & 3 \\ 5 & -1 \end{bmatrix} \cdot \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Step 3. Check

Experiment 1: $\mathbf{R}\,[1\ 1]^T = [5\ 5]^T$. Experiment 2: $\mathbf{R}\,[1\ -1]^T = [3\ -1]^T$. ✓

7. The power of a two-port

Let P_N be the sum of the powers absorbed by all elements inside the two-port N.

Theorem

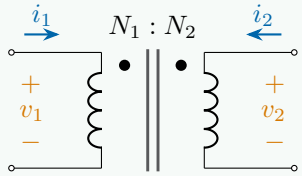
$$P_N = v_1\, i_1 + v_2\, i_2.$$

Why. Connect sources to the two ports. By Tellegen's theorem (previously), the sum of the absorbed powers of all branches is zero. The port sources absorb $-v_1\, i_1$ and $-v_2\, i_2$, so the elements inside N absorb $v_1\, i_1 + v_2\, i_2$.

Passive and active two-ports

A two-port is **passive** if $P_N \geq 0$ for all possible port variables (v_1, i_1, v_2, i_2). It is **active** if it is not passive.

Example 14.8 — is the ideal transformer passive?



Find the power P_N of the ideal transformer. Is it passive or active?

Solution

Step 1. The power

Plug $v_1 = n\, v_2$ and $i_2 = -n\, i_1$ into the theorem:

$$P_N = v_1\, i_1 + v_2\, i_2 = n\, v_2\, i_1 - n\, v_2\, i_1 = 0.$$

Step 2. Passive or active

$P_N = 0 \geq 0$ for all port variables, so the ideal transformer is passive. It neither absorbs nor delivers power: what enters one port leaves the other.

Summary

Matrix	Equation	How to find it
resistance $\mathbf{R}$	$[\,v_1\ v_2\,]^T = \mathbf{R}\,[\,i_1\ i_2\,]^T$	KVL, or open-circuit tests ($i_2 = 0$, then $i_1 = 0$)
conductance $\mathbf{G}$	$[\,i_1\ i_2\,]^T = \mathbf{G}\,[\,v_1\ v_2\,]^T$	short-circuit tests; $\mathbf{G} = \mathbf{R}^{-1}$ when both exist
hybrid $\mathbf{H}$, $\mathbf{H}'$	$[\,v_1\ i_2\,]^T = \mathbf{H}\,[\,i_1\ v_2\,]^T$	port 2 shorted, then port 1 open; $\mathbf{H}' = \mathbf{H}^{-1}$
transmission $\mathbf{T}$	$[\,v_1\ i_1\,]^T = \mathbf{T}\,[\,v_2\ -i_2\,]^T$	port 1 in terms of port 2
power	$P_N = v_1\,i_1 + v_2\,i_2$	passive if $P_N \geq 0$ for all port variables

A. Passivity from the matrix

A two-port is passive if $P_N \geq 0$ for all port variables. When $\mathbf{R}$, $\mathbf{G}$ or $\mathbf{H}$ exists, this can be decided from the matrix alone.

A.1. Power as a quadratic form

Write $\mathbf{i} = [\, i_1 \;\; i_2 \,]^T$ and $\mathbf{v} = [\, v_1 \;\; v_2 \,]^T$. With $\mathbf{v} = \mathbf{R}\,\mathbf{i}$,

$$P_N = v_1\, i_1 + v_2\, i_2 = \mathbf{i}^T\mathbf{v} = \mathbf{i}^T\mathbf{R}\,\mathbf{i} = r_{11}\, {i_1}^2 + (r_{12} + r_{21})\, i_1\, i_2 + r_{22}\, {i_2}^2.$$

Only the sum $r_{12} + r_{21}$ appears. So only the **symmetric part** of $\mathbf{R}$ matters:

$$\mathbf{R}_s = \tfrac{1}{2}\,(\mathbf{R} + \mathbf{R}^T) = \begin{bmatrix} r_{11} & \tfrac{1}{2}(r_{12} + r_{21}) \\ \tfrac{1}{2}(r_{12} + r_{21}) & r_{22} \end{bmatrix}, \qquad P_N = \mathbf{i}^T\mathbf{R}_s\,\mathbf{i}.$$

A.2. The test

Passivity test

The two-port is passive if and only if $\mathbf{R}_s$ is positive semidefinite. For a 2×2 matrix this means

$$r_{11} \geq 0, \qquad r_{22} \geq 0, \qquad r_{11}\, r_{22} - \tfrac{1}{4}\,(r_{12} + r_{21})^2 \geq 0.$$

If one condition fails, some port currents give $P_N < 0$: the two-port is active.

The same works with the other matrices. With $\mathbf{G}$, $P_N = \mathbf{v}^T\mathbf{G}\,\mathbf{v}$: test $\frac{1}{2}(\mathbf{G} + \mathbf{G}^T)$. With $\mathbf{H}$, take $\mathbf{x} = [\, i_1 \;\; v_2 \,]^T$; then $[\, v_1 \;\; i_2 \,]^T = \mathbf{H}\,\mathbf{x}$ and $P_N = \mathbf{x}^T\mathbf{H}\,\mathbf{x}$: test $\frac{1}{2}(\mathbf{H} + \mathbf{H}^T)$. The transmission matrix $\mathbf{T}$ does not give such a form, because it relates port 1 to port 2 instead of the voltages to the currents.

Example A.1 — the two-port of Example 14.4

$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$. It contains a dependent source. Is it passive?

Solution

Step 1. The symmetric part

$$\mathbf{R}_s = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Step 2. The test

$3 \geq 0$, $1 \geq 0$ and $3 \cdot 1 - 0^2 = 3 \geq 0$. The two-port is **passive**: a dependent source inside does not make a two-port active by itself.

Example A.2 — the ideal transformer

$\mathbf{H} = \begin{bmatrix} 0 & n \\ -n & 0 \end{bmatrix}$ (Example 14.5). Is it passive?

Solution

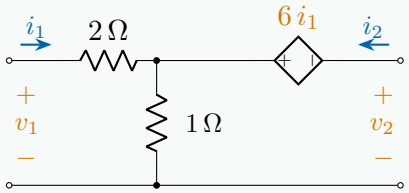
Step 1. The symmetric part

$\mathbf{H}$ is antisymmetric, $\mathbf{H}^T = -\mathbf{H}$, so $\frac{1}{2}(\mathbf{H} + \mathbf{H}^T) = \mathbf{0}$.

Step 2. The test

$P_N = 0$ for all port variables: the transformer is **passive** and absorbs no power, as in Example 14.8.

Example A.3 — a larger gain



The circuit of Example 14.2 with the dependent source $6\,i_1$ instead of $2\,i_1$. Is it passive?

Solution

Step 1. The matrix

As in Example 14.2, with 2 replaced by 6:

$$\mathbf{R} = \begin{bmatrix} 3 & 1 \\ 1-6 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -5 & 1 \end{bmatrix}, \qquad \mathbf{R}_s = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}.$$

Step 2. The test

$3 \cdot 1 - (-2)^2 = -1 < 0$: the two-port is **active**.

Step 3. A check

Take $i_1 = 1\,\text{A}$ and $i_2 = 2\,\text{A}$:

$$P_N = 3 \cdot 1^2 + (1-5) \cdot 1 \cdot 2 + 1 \cdot 2^2 = -1\,\text{W}.$$

The two-port delivers 1 W to the outside. ✓